

# Adding and Subtracting in Scientific Notation

## Lesson Plan

### 1. Lesson Information

|        |                                                          |              |                                            |
|--------|----------------------------------------------------------|--------------|--------------------------------------------|
| Course | Grade 7 Accelerated (1205050)                            | Unit         | 4 — Solving Problems with Rational Numbers |
| Lesson | Lesson 1 — Adding and Subtracting in Scientific Notation | Date         | <hr/>                                      |
| Length | 53 minutes (one period)                                  | Benchmark(s) | MA.8.NSO.1.5                               |

### 2. Standards

**MA.8.NSO.1.5** Add, subtract, multiply and divide numbers expressed in scientific notation with procedural fluency.

**Must.** Add and subtract numbers written in scientific notation with procedural fluency, and write every answer with a coefficient at least 1 and less than 10.

**Must not.** The two exponents stay within 2 of each other — that is this benchmark's own limit. Multiplication and division are 4.02–03; significant digits are 4.04.

### 3. Mathematical Thinking and Reasoning Standards

*the evidence for each, from this period*

| MTR            | What it asks                                                                        | The evidence in this lesson                                                                                                                               |
|----------------|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| MA.K12.MTR.5.1 | Use patterns and structure to help understand and connect mathematical concepts.    | Notes I — the powers of ten are set against the place-value names, so 'the powers must match' is read off the structure rather than announced as a rule.  |
| MA.K12.MTR.6.1 | Assess the reasonableness of solutions.                                             | Boards 3 and 4 — the coefficient is checked against $1 \leq a < 10$ before the board goes up, in both directions.                                         |
| MA.K12.MTR.4.1 | Engage in discussions that reflect on the mathematical thinking of self and others. | Board 6 — when the room splits between $1.23 \times 10^7$ and $12.3 \times 10^6$ , which have the same value, the sixty-second re-vote before the reveal. |

## 4. Learning Target and Essential Question

**Learning target.** I can add and subtract two numbers written in scientific notation, and write the answer in scientific notation.

**Essential question.** Why must two numbers written in scientific notation carry the same power of ten before they can be added?

**Building on.** MA.8.NSO.1.4 (3.08–3.09) — scientific notation, the coefficient rule and reading notation back; Grade 6 — adding and subtracting decimals by place value.

**Working toward.** 4.02–03 — multiplying and dividing in scientific notation; MA.8.NSO.1.6 (4.04) — real-world problems and significant digits.

## 5. Sequence

*read from the deck; never typed*

| Segment                             | Min       | What the teacher does                                                                        | What students do                           |
|-------------------------------------|-----------|----------------------------------------------------------------------------------------------|--------------------------------------------|
| Settle in; post the learning target | 1         | Post the target; say the one line the Teacher's Edition gives for it, and nothing else yet.  | Copy the target.                           |
| Warm-Up                             | 5         | Two minutes silent, then reveal. One sentence of reteach per question at most.               | Four retrieval questions, alone, no notes. |
| Notes                               | 11        | Teach from the front. Expand once per law; say the OFF line out loud.                        | Copy the notes; answer the checks aloud.   |
| Example 1                           | 3         | Work the example; name the wrong boards before they appear.                                  | Watch, then work the Your Turn alone.      |
| Your Turn                           | 3         | Circulate; reveal and name what a wrong board most likely did.                               | Work it alone; boards up on the cue.       |
| Example 2                           | 3         | Work the example; name the wrong boards before they appear.                                  | Watch, then work the Your Turn alone.      |
| Your Turn                           | 3         | Circulate; reveal and name what a wrong board most likely did.                               | Work it alone; boards up on the cue.       |
| Whiteboards                         | 13        | One question at a time. Boards down until the cue, all up together; scan the back row first. | Answer on a board; hold it up on three.    |
| Independent Set                     | 6         | Silent. Circulate and mark what you see; do not teach.                                       | Six questions, written, alone.             |
| IXL                                 | 5         | Assign the day's IXL skills.                                                                 | Start the skills; finish tonight.          |
| <b>Total</b>                        | <b>53</b> |                                                                                              |                                            |

## 6. Gradual Release

**I do.** The Notes slides and the worked half of each Example: the teacher writes, the class copies, and the OFF line is said out loud rather than printed.

**We do.** Each Example's Your Turn: the same steps on the student's own numbers, revealed and named a minute later.

**You do.** The nine-question whiteboard round (13 minutes) and then the six-question independent set, silent and written.

## 7. Higher-Order Questions

| Question                                                                                                                              | DOK   |
|---------------------------------------------------------------------------------------------------------------------------------------|-------|
| Why can $5.2 \times 10^6$ and $3.1 \times 10^6$ be added straight away, while $5.2 \times 10^6$ and $3.1 \times 10^5$ cannot?         | DOK 2 |
| Devon adds the exponents and gets $10^6$ . What would have to be true about the expression for adding exponents to be the right move? | DOK 3 |
| $12.3 \times 10^6$ and $1.23 \times 10^7$ are the same number. Why does one of them count as an answer and the other does not?        | DOK 3 |

## 8. Checks for Understanding

*each board, and what to do about it*

| # | What the board asks                                                                                                                                            | Answer                  | If the board is wrong                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | $(3.2 \times 10^5) + (4.1 \times 10^5)$                                                                                                                        | $7.3 \times 10^5$       | $7.3 \times 10^{10}$ — the exponents were added as well as the coefficients [B1G-M 8.NSO.1.5 misconception: exponents added]; $1.312 \times 10^{11}$ — the numbers were multiplied [8.NSO.1.5: the operation is addition]                                                                                                                                                                                                                              |
| 2 | A stadium sold $4.2 \times 10^4$ tickets on Friday and $3.5 \times 10^4$ on Saturday. Each ticket cost 38 dollars. How many tickets were sold on the two days? | $7.7 \times 10^4$       | anything built from 38 — the unneeded figure was used [the question asks for a count, not a cost]; $7.7 \times 10^8$ — the exponents were added [B1G-M 8.NSO.1.5 misconception]                                                                                                                                                                                                                                                                        |
| 3 | $(8 \times 10^{-6}) + (3 \times 10^{-6})$                                                                                                                      | $1.1 \times 10^{-5}$    | $11 \times 10^{-6}$ — right value, but 11 is not less than 10 [B1G-M 8.NSO.1.5 named misconception: leaving $12 \times 10^9$ instead of $1.2 \times 10^{10}$ ]; $1.1 \times 10^{-11}$ — the exponents were added, giving $11 \times 10^{-12}$ , and the answer was then finished                                                                                                                                                                       |
| 4 | $(5.6 \times 10^8) - (4.9 \times 10^8)$                                                                                                                        | $7 \times 10^7$         | $0.7 \times 10^8$ — right value, but 0.7 is not at least 1 [8.NSO.1.5: the coefficient satisfies $1 \leq a < 10$ ]; $1.05 \times 10^9$ — added instead of subtracting                                                                                                                                                                                                                                                                                  |
| 5 | $(1.6 \times 10^4) + (2.3 \times 10^6)$ . Which is the sum?                                                                                                    | $C — 2.316 \times 10^6$ | (A) the coefficients were added without matching the powers of ten first [8.NSO.1.5: the powers must match]; (B) the coefficients were added and the exponents were added [B1G-M 8.NSO.1.5 named misconception: $(1.3 \times 10^3) + (3.4 \times 10^5) = 4.7 \times 10^8$ ]; (D) $1.6 \times 10^4$ was rewritten as $0.16 \times 10^6$ — the decimal point moved one place too few [8.NSO.1.5: the point moves as many places as the exponent changes] |
| 6 | $(2.5 \times 10^6) + (9.8 \times 10^6)$ . Which is the sum, written in scientific notation?                                                                    | $D — 1.23 \times 10^7$  | (A) equal in value, but 12.3 is not less than 10 [B1G-M 8.NSO.1.5 named misconception: leaving $12 \times 10^9$ instead of $1.2 \times 10^{10}$ ]; (B) the exponents were added as well as the coefficients [8.NSO.1.5: the power of ten does not change]; (C) the coefficient was repaired but the exponent was not raised [8.NSO.1.5: point one place left, exponent up one]                                                                         |
| 7 | $(7.8 \times 10^{-2}) - (6.4 \times 10^{-3})$                                                                                                                  | $7.16 \times 10^{-2}$   | $1.4 \times 10^{-2}$ — the coefficients were subtracted without matching the powers [8.NSO.1.5]; $7.16 \times 10^{-5}$ — the exponents were added, $-2 + (-3)$ [8.NSO.1.5: the power of ten is a label, not a term]                                                                                                                                                                                                                                    |

| # | What the board asks                                                                                                               | Answer              | If the board is wrong                                                                                                                                                        |
|---|-----------------------------------------------------------------------------------------------------------------------------------|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 8 | $(4.5 \times 10^6) + (n \times 10^6) = 1.02 \times 10^7$ .<br>What is the value of n?                                             | $n = 5.7$           | -3.48 — the target was not rewritten over $10^6$ before subtracting [8.NSO.1.5: match the powers first]; $5.7 \times 10^6$ — the blank holds a coefficient, not a whole term |
| 9 | Devon writes $(1.9 \times 10^2) + (6.6 \times 10^4) = 8.5 \times 10^6$ . Name the two mistakes he made, and give the correct sum. | $6.619 \times 10^4$ | 'he added wrong' — no mistake is actually named [the item asks which two]; $8.5 \times 10^4$ — the exponent mistake was caught and the matching mistake was not              |

**MTR.4.1 — the one peer move.** When a board splits — no option holding about two-thirds of the room — sixty seconds of “convince the person next to you”, then re-vote, then reveal. The minute comes out of the whiteboard remainder.

## 9. Differentiation

**ESE / IEP.** Give the place-value strip from Notes I with the power written above each name, and let the student write both numbers as words first — 'four point nine five ten-thousands' and 'twelve point one ten-thousands' — before any symbol is written. The matching step becomes reading, not procedure.

**ELL.** Coefficient and power of ten are the two phrases every question uses. Write  $7.7 \times 10^4$  once with COEFFICIENT under the 7.7 and POWER OF TEN under the  $10^4$ , and point at the labels on every board. Add or subtract is the third phrase: read each board's operation aloud before the boards go up.

**Enrichment.** Ask for  $(4.2 \times 10^6) + (3.9 \times 10^5) + (1.1 \times 10^4)$  — three terms, all within two of each other — and then why the benchmark stops at a gap of two.

## 10. Closure, Vocabulary, Materials, Practice

**Closure.** Board 9 is written work and it is the exit evidence: the student has to name both of Devon's mistakes separately and give  $6.619 \times 10^4$ . Read the boards, not the papers.

**Vocabulary.** **scientific notation; coefficient; common power of ten**

**Materials.** Whiteboards and markers. Calculators are allowed on everything; entering  $8\text{E}-6 + 3\text{E}-6$  and reading  $1.1\text{E}-5$  back off the display is worth thirty seconds at whiteboard 3.

**Practice.** The six-question independent set is worked silently in class; whatever is not finished goes home. IXL, all skills required to a SmartScore of 67, due at the start of the next class: Add and subtract numbers written in scientific notation — HUR.